# Use the force diagram to find acceleration

**GUIDED NOTES / use with the projected diagram** Name _____

## A SAY THE WORD

### Component / ES: componente / PT: componente

The downhill part of weight. A way to calculate — not an extra force.

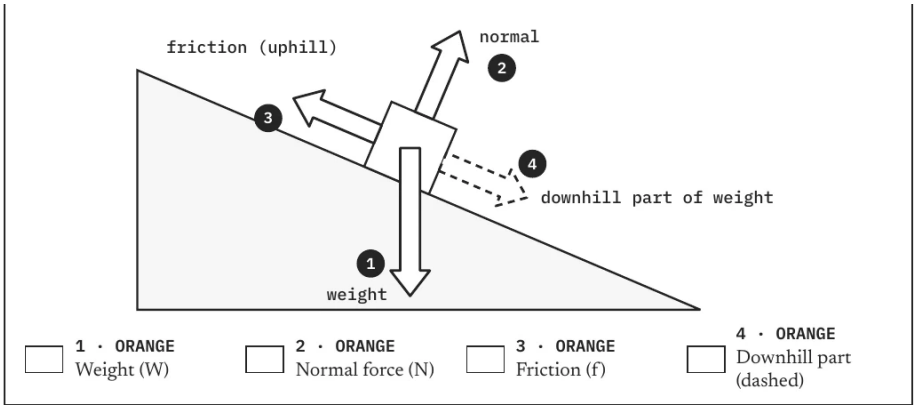
Also on your note page: Normal force, Weight, Acceleration direction.

## B READ THE RELATIONSHIP

### A component is the part of one vector along one chosen axis.

Resolve one force along two perpendicular axes

## C TRACE THE ORIGINAL FIGURE




Use the numbered key. Trace one feature. Tell your partner what it means.

## D FOLLOW THE WORKED EXAMPLE

### A 2 kg block has a 7 N downhill weight component and 3 N of friction uphill. Find its acceleration.

**Given** m = 2 kg 7 N downhill 3 N uphill

**Rule** a = F net ÷ m

**Work** F net = 7 − 3 = 4 N downhill a = 4 ÷ 2

**Answer** 2 m/s² downhill

## E SAVE FOR THE FINAL MEETING

From the diagram, the net force gives XJ's acceleration and its direction. Get the net force's direction right and you know which way its path will bend.

# Read the physics: Component

SUPPORTED READING / read one part, then point to the evidence Name _____

## 01 PICTURE IT

On a ramp, gravity still points straight down. The normal force points straight out of the surface. Choose one axis along the ramp and one perpendicular to it.

## 02 HOW IT WORKS

Split weight into two parts. The parallel part pulls downhill; the perpendicular part presses into the ramp. Calculate with the parts or with the original weight, never with both together.

## 03 WHAT TO CHECK

Subtract uphill friction from the downhill force component. If the block stays on the ramp, its acceleration perpendicular to the ramp is zero.

## 04 TRY THE EXAMPLE FROM YOUR NOTES

A 2 kg block has a 7 N downhill weight component and 3 N of friction uphill. Find its acceleration. 2 m/s² downhill.

## 05 CONNECT TO XJ

From the diagram, the net force gives XJ's acceleration and its direction. Get the net force's direction right and you know which way its path will bend.

### SAY IT BEFORE YOU WRITE

Why would adding both weight and its components be a mistake? Along the ramp I use ___. Across the ramp I use ___.

# Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

## Normal force

ES: fuerza normal / PT: força normal

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain normal force to a friend?<br>In my words, this means ___. | A crate rests on a ramp. Which way does the ramp push the crate?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| How would you draw the normal force on a book on a level table?<br>My example shows ___. | Someone draws the normal force along the ramp. What would you rotate?<br>I would change ___ because ___. |

## Weight

ES: peso / PT: peso

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain weight to a friend?<br>In my words, this means ___. | When a book rests on a table, which object supplies its weight?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| Using 10 N/kg, what arrow represents the weight of a 3 kg book?<br>My example shows ___. | A book slides down a ramp. Should its whole weight arrow point along the ramp?<br>I would change ___ because ___. |

# Frayer models

**SECTION A / VOCABULARY** Name _____

Say your idea in any language. Write, draw, or label it in the box.

## Component

ES: componente / PT: componente

| **Meaning in my words** | **How can you tell?** |
|---|---|
| How would you explain component to a friend?<br>In my words, this means ___. | Which part of the block's weight pulls it down the ramp?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| How could you split one downward weight arrow using ramp axes?<br>My example shows ___. | Should the downhill component be added to the full weight as another force?<br>I would change ___ because ___. |

## Acceleration direction

ES: sentido de la aceleración / PT: sentido da aceleração

| **Meaning in my words** | **How can you tell?** |
|---|---|
| How would you explain acceleration direction to a friend?<br>In my words, this means ___. | A cart rolls uphill on a frictionless ramp. Which way does it accelerate?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| At the top of its trip, draw the acceleration arrow just as the cart stops.<br>My example shows ___. | "The cart moves uphill, so acceleration points uphill." Which arrows got mixed up?<br>I would change ___ because ___. |

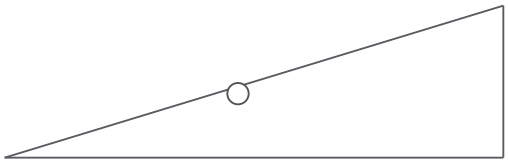
# Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

## Draw the real weight once, then draw its two components in a separate box. Circle the component along the ramp.

Labels you can use: parallel · perpendicular · downhill.

Real forces



Components

**Tell a partner**

Along the ramp I use ___. Across the ramp I use ___.

# One pull, two components

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SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

**Our goal:** Represent one force without counting it twice.

**CHOOSE ONE JOB TOGETHER**

**NAVIGATION ENGINEER** ☐

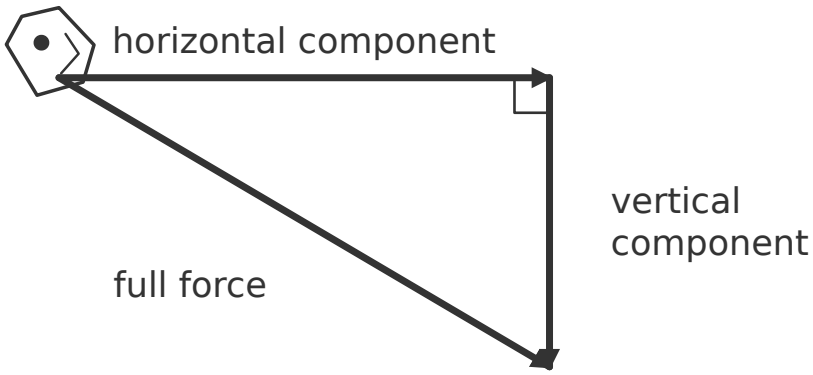
Prepare the component view for the calculator. Mark the full-force arrow that these components replace.

**MAP GRAPHICS EDITOR** ☐

Show the full force and its component view. Write a caption that prevents counting the same pull twice.

**Word help:** force / component / sum

A navigation graphic lists a gravitational force AND its two components as three separate pulls on XJ.




## 01 OUR FIRST CALL

**Which arrows should enter the force sum? Explain your choice.**

Use ___ because the components represent ___.

## 02 OUR REVISED CALL

**Wait for the reveal. Keep your first call.**

New clue: _____

Check the calculator request. What correction would you send before it runs?

Now we would say ___ because ___.

**Switch speakers.** Are you counting one pull twice?

| ☐ GREEN | ☐ RED | ☐ YELLOW | ☐ BLUE | ☐ PURPLE | ☐ ORANGE | ☐ PENCIL |
|---|---|---|---|---|---|---|
| speeding up | slowing down | steady / balanced | position / displacement | velocity | forces | other marks |